

FarmFriend

Gateway to Modern Farming

Team ReBuggers

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Course Details:

Software Engineering Laboratory

Course Code: CSE 3422

Section: D

Project Code Link:

FarmFriend Github Repository

Introduction

FarmFriend is a web-based full-stack agricultural support system that digitizes and modernizes traditional farming methods in Bangladesh. Because it creates jobs for a large number of people and guarantees food security nationwide, the agricultural sector is the foundation of the national economy. Due to a lack of modern decision-making systems and adequate access to technology, the agricultural sector still relies on manual labor.

FarmFriend establishes a solution through its combination of software engineering methods and agricultural problem-solving techniques. The platform uses real-time data processing together with automated systems and intelligent decision support to help farmers make accurate agricultural decisions at the right times. The system delivers a complete digital solution to Bangladeshi farmers through its features which include crop recommendation, weather monitoring, equipment sharing, and expert consultation.

The project follows essential software engineering principles which include modularity, scalability, maintainability, and reusability for its design work. FarmFriend uses Django and React and RESTful APIs and automated testing tools to achieve system reliability and performance and long-term system sustainability. The system shows which technological tools can transform agricultural productivity and efficiency within developing nations such as Bangladesh.

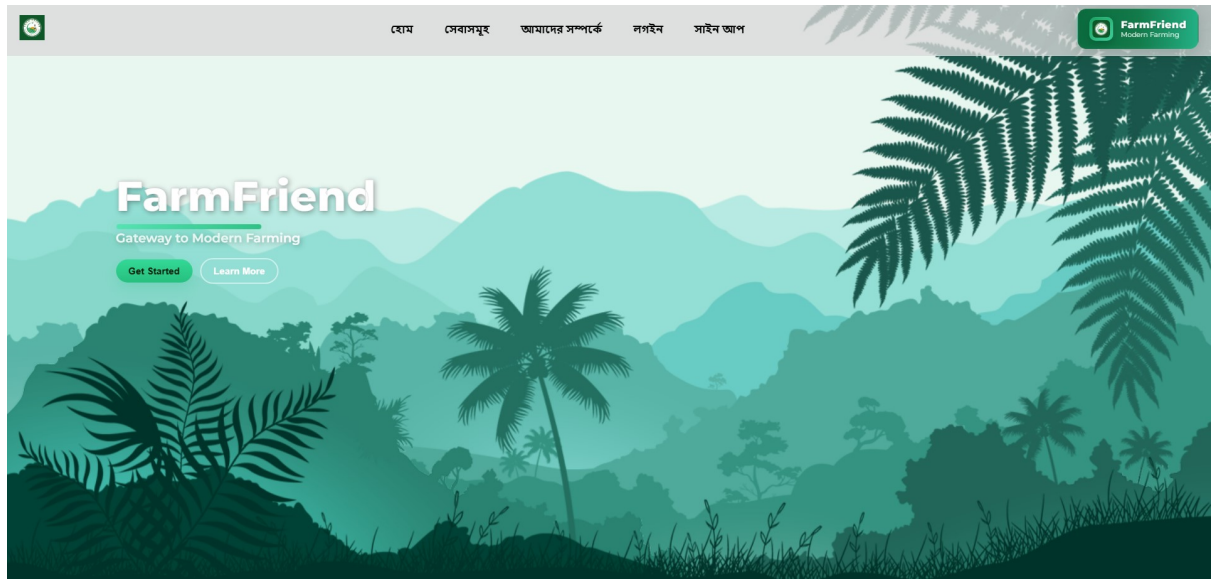


Figure 1: FarmFriend

Objectives

The primary objective of FarmFriend is to develop a robust and scalable agricultural support system that addresses real-world challenges faced by farmers. The specific objectives of the project are as follows:

- To design and implement a full-stack web application using modern software engineering practices.

- To provide farmers with data-driven decision-making tools for crop selection, weather awareness, and resource management.
- To ensure modular and maintainable system architecture using Django and React frameworks.
- To integrate automated testing mechanisms to improve software reliability and reduce system errors.
- To enhance usability through a simple, intuitive, and farmer-friendly user interface.
- To promote efficient use of agricultural resources and reduce operational costs through digital solutions.
- To demonstrate the practical application of software engineering concepts in solving real-life problems in Bangladesh.

Motivation

Agriculture remains the backbone of Bangladesh's economy, yet a large portion of farmers still depend on traditional knowledge and manual practices. Factors such as unpredictable weather patterns, lack of timely information, limited access to expert guidance, and inefficient resource utilization often lead to low productivity and financial loss. Moreover, small-scale farmers face challenges in accessing modern tools, storage facilities, and reliable market information.

The motivation behind developing FarmFriend stems from the need to introduce a technological revolution in the agricultural sector of Bangladesh. With the rapid growth of internet connectivity and smartphone usage in rural areas, there is a strong opportunity to leverage software-based solutions to empower farmers. FarmFriend was envisioned as a digital companion that simplifies farming decisions through automation, data analysis, and real-time updates.

By incorporating modern web technologies and software engineering principles, the project aims to reduce the knowledge gap between traditional farming practices and modern agricultural techniques. The platform has the potential to improve crop yield, minimize losses, increase farmer income, and promote sustainable agricultural practices. In the long run, FarmFriend can contribute significantly to the digital transformation of agriculture in Bangladesh, supporting national goals such as food security, economic growth, and technological innovation.

Project Features

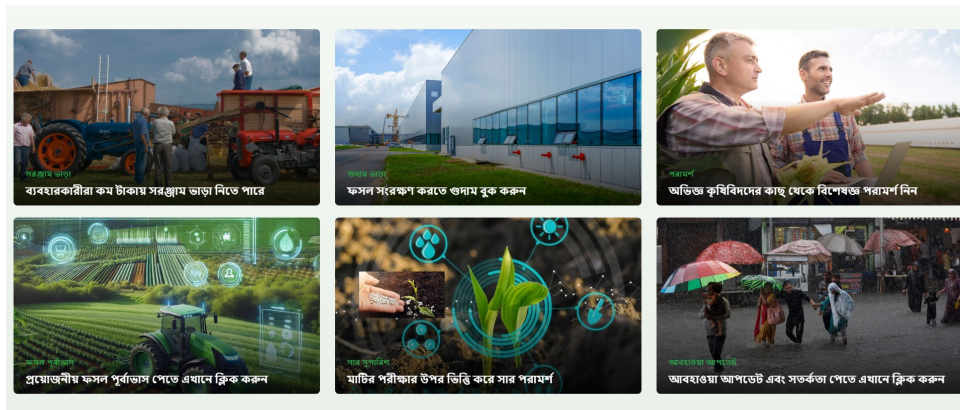


Figure 2: FarmFriend Services



Figure 3: FarmFriend Services

1. Multi-User Dashboard System

• Farmer Dashboard:

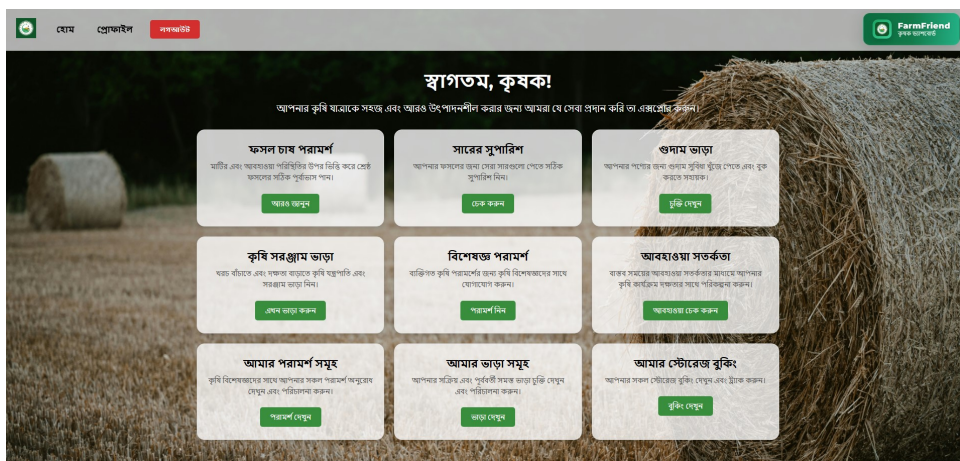


Figure 4: Farmer Dashboard

- Provides access to crop recommendations, weather updates, fertilizer suggestions, and equipment rental.
- Allows farmers to track storage bookings, consult experts, and manage farming activities from a single interface.

• Agronomist Dashboard:

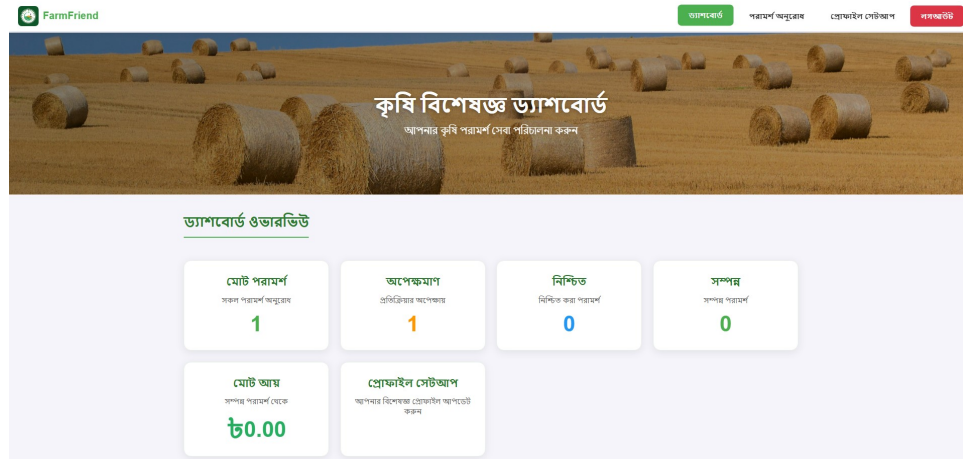


Figure 5: Agronomist Dashboard

- Enables agronomists to view farmer requests and provide expert consultations.
- Allows sharing of crop advice, pest control guidance, and soil management recommendations.

• Storage Owner Dashboard:

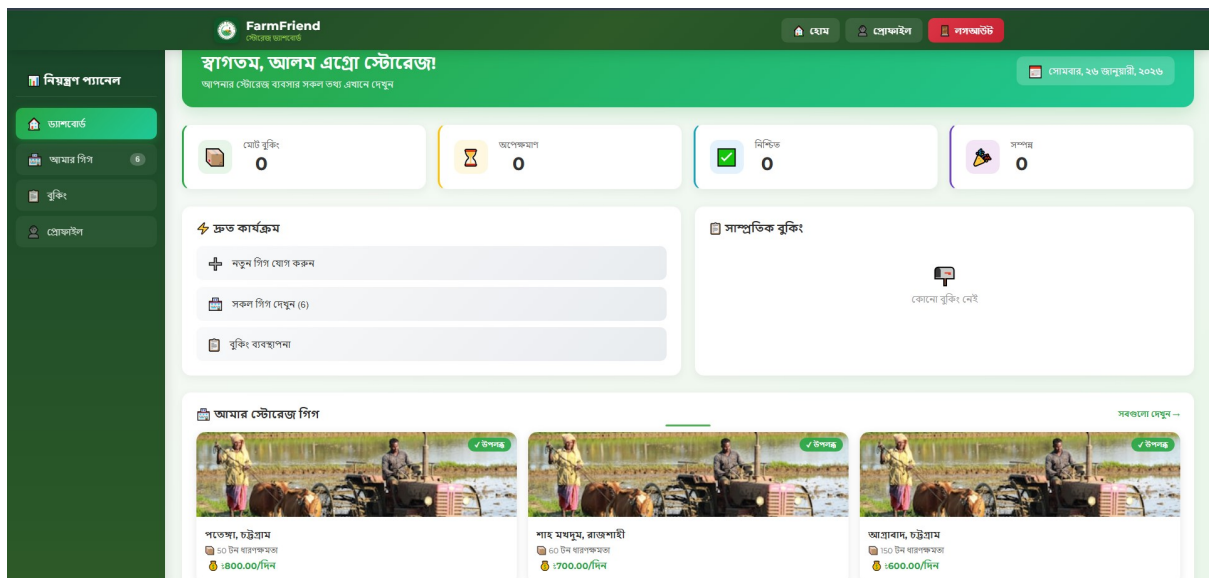


Figure 6: Storage Owner Dashboard

- Allows storage owners to manage available storage capacity and booking requests.

- Helps track inventory status and reduce post-harvest losses through efficient storage management.

- **Equipment Renter Dashboard:**

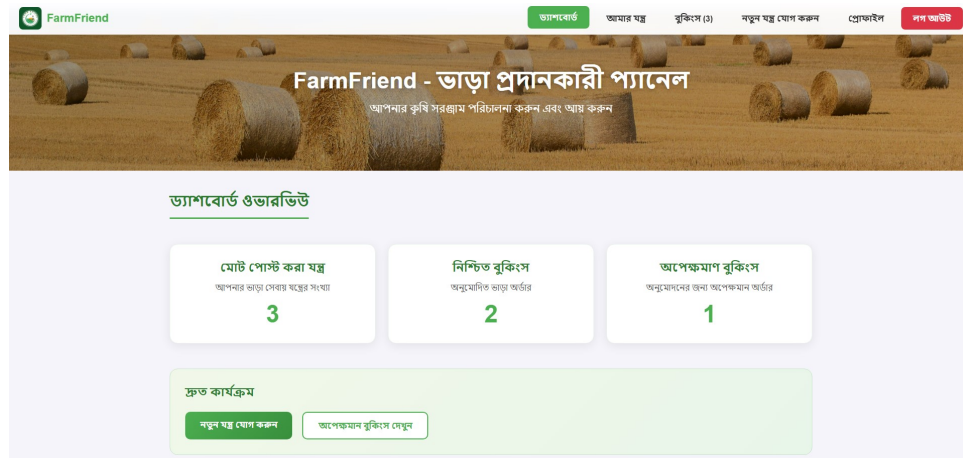


Figure 7: Renter Dashboard

- Enables equipment owners to list farming tools for rent.
- Allows management of rental requests, availability schedules, and pricing.

2. Crop Recommendation System

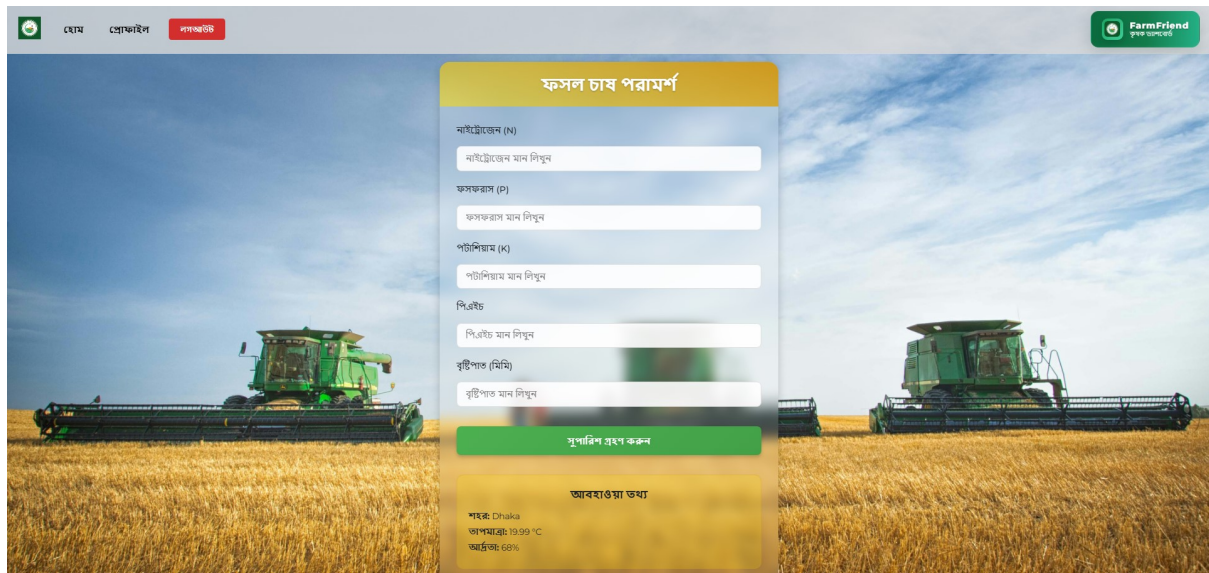


Figure 8: Feature Overview

- Uses agricultural data to recommend suitable crops.
- Helps farmers choose crops based on soil and climate conditions.
- Improves productivity and reduces crop failure risks.

3. Weather Monitoring System



Figure 9: Feature Overview

- Provides real-time weather updates.
- Alerts farmers about rainfall, storms, and extreme conditions.
- Helps in planning irrigation and harvesting.

4. Equipment Sharing Platform

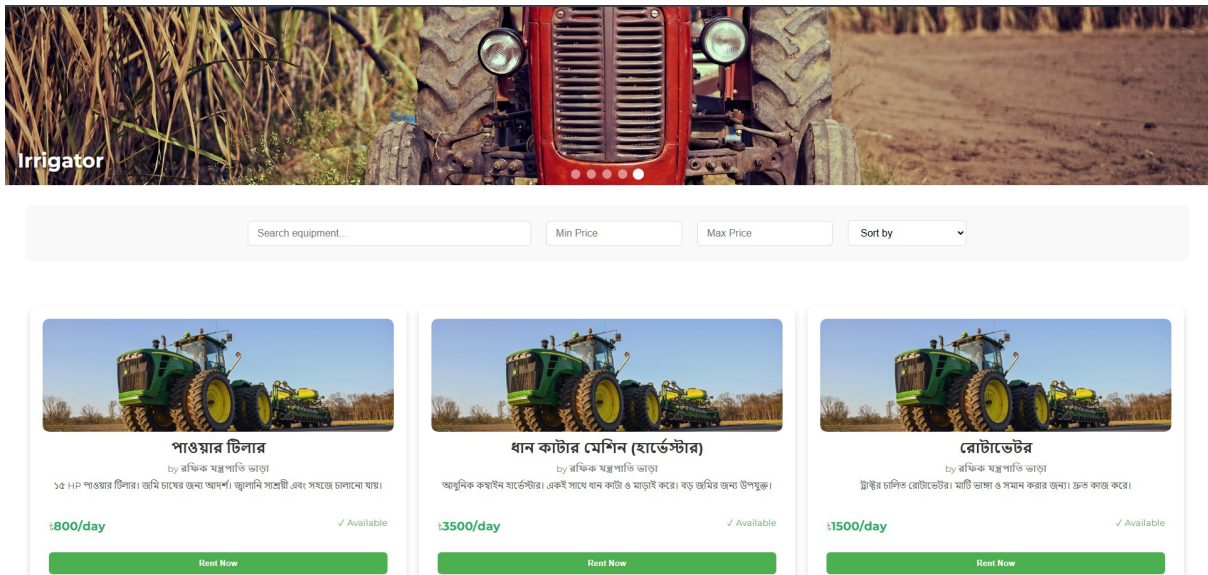


Figure 10: Feature Overview

- Allows farmers to rent or share farming tools.
- Reduces cost of ownership.

- Encourages community-based resource utilization.

5. Expert Consultation Module

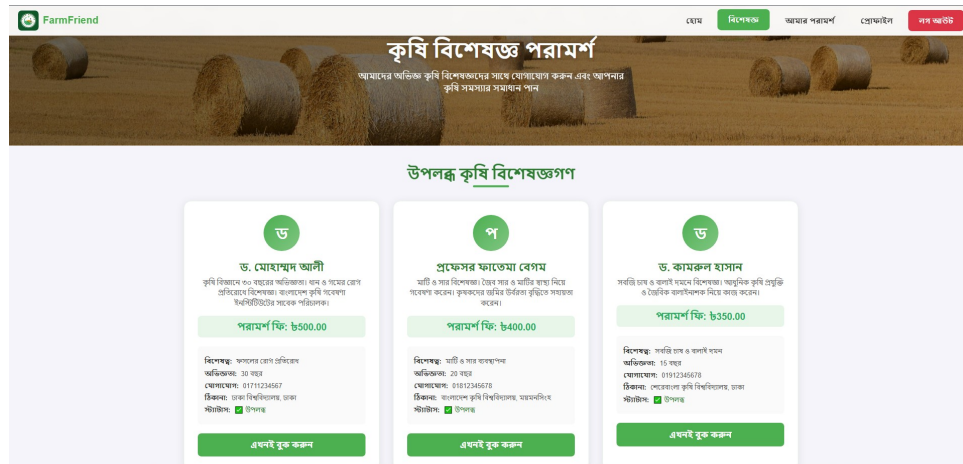


Figure 11: Feature Overview

- Enables communication with agricultural experts.
- Provides solutions for crop disease and soil management.
- Supports informed decision-making.

6. Storage Facility Booking

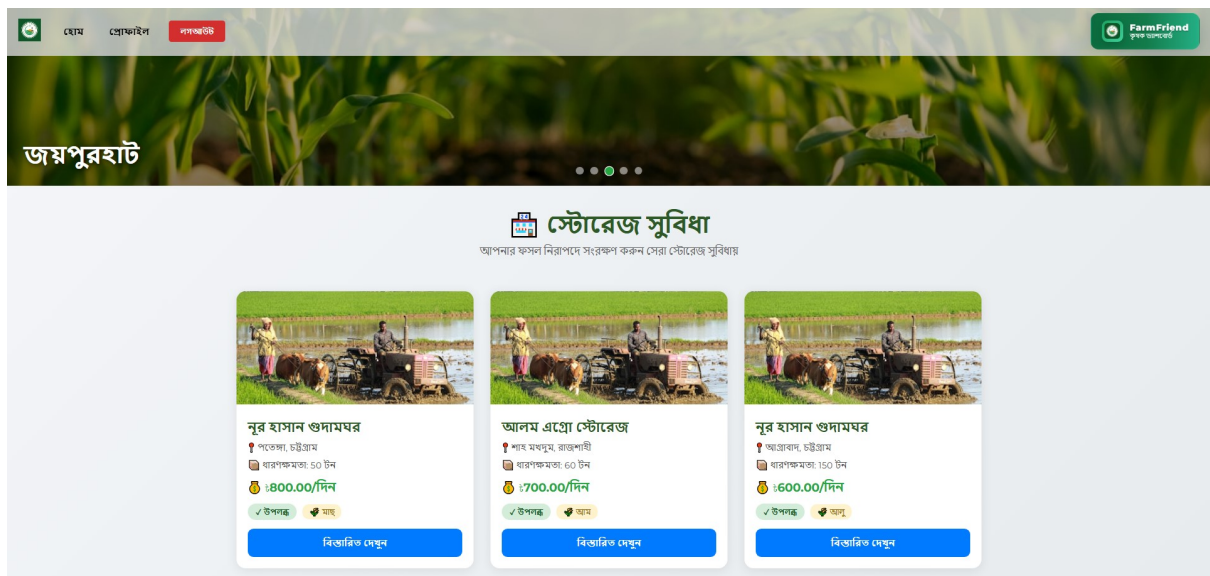


Figure 12: Feature Overview

- Allows booking of storage facilities.
- Reduces post-harvest loss.
- Maintains inventory records.

7. Fertilizer Recommendation System

The screenshot displays the 'সার সুপারিশ ফর্ম' (Fertilizer Recommendation Form) within the FarmFriend application. The form is overlaid on a background image of a hand holding soil. The form includes a header section for 'আবহাওয়া তথ্য' (Weather Information) with fields for 'শহর: Dhaka', 'আমের তাপমাত্রা: 19.99 °C', 'আর্দ্রতা: 60%', and 'অবস্থা: haze'. Below this, there are input fields for 'নাইট্রোজেন' (Nitrogen), 'ফসফরাস' (Phosphorus), 'পটাশিয়াম' (Potassium), 'অম্লীয়তা' (Acidity), and 'অন্যান্য তথ্য' (Other Information). A dropdown menu for 'অন্যান্য তথ্য' is currently set to 'ভাল' (Good). At the bottom, there is a green button labeled 'সারের সুপারিশ গ্রহণ করুন' (Get Fertilizer Recommendation).

Figure 13: Feature Overview

- Suggests fertilizers based on crop and soil data.
- Helps optimize fertilizer usage.
- Supports sustainable farming.

Technology Used



React.js was used for frontend development due to its component-based architecture and efficient state management. It enables the creation of dynamic, responsive, and reusable user interface components, ensuring a smooth user experience across different devices.



Django was selected as the backend framework because of its robust architecture, built-in security features, and scalability. It simplifies backend development by handling URL routing, business logic, and server-side operations efficiently.



MySQL was used as the database management system due to its reliability, structured data handling, and strong support for relational data. It efficiently manages user data, farming records, bookings, and transactional information.



Django Authentication was implemented to manage secure user registration, login, and role-based access control. It ensures that different users such as farmers, agronomists, storage owners, and equipment renters can securely access authorized system features.



RESTful APIs were developed to enable seamless communication between the React frontend and Django backend. APIs allow efficient data exchange, system modularity, and easier future expansion such as mobile application integration.



Playwright was used for automated end-to-end testing to verify system functionality, user interactions, and cross-browser compatibility. Automated testing improves reliability and reduces manual testing effort.



Git and GitHub were used for version control and team collaboration. GitHub enabled code sharing, change tracking, issue management, and collaborative development among team members.

Automated Testing

Automated testing was implemented using **Playwright** to ensure the reliability, correctness, and stability of the system. End-to-end testing was chosen to simulate real user interactions and validate critical workflows across different user roles within the platform.

The automated test script verifies the complete rental workflow involving multiple users and includes the following scenarios:

- **Renter Workflow:**

- Creation of a renter account through the signup process.
- Authentication by logging in as a renter.

- Posting an agricultural item for rent with relevant details.
- Logging out of the system.
- **Farmer Workflow:**
 - Creation of a farmer account through the signup process.
 - Authentication by logging in as a farmer.
 - Navigation to the rental marketplace.
 - Renting the previously posted item.
 - Logging out of the system.

The test cases also validate user authentication, role-based access control, form submissions, and navigation between system pages. By automating these critical workflows, the system ensures consistent behavior, reduces regression issues, and minimizes manual testing effort. The complete Playwright testing script is available in the project’s GitHub repository.

Challenges

During the development of FarmFriend, several technical and practical challenges were encountered. These challenges reflect real-world constraints faced while building a software system tailored for the agricultural sector in Bangladesh.

- **Frontend and Backend Integration:** Integrating the React frontend with the Django backend required careful handling of RESTful APIs, authentication tokens, and data synchronization. Ensuring seamless communication between the client and server was a significant challenge.
- **Asynchronous API Communication:** Managing asynchronous data fetching and handling API responses in the frontend required proper state management to avoid data inconsistency and UI delays.
- **Designing the System in Bengali:** Designing the entire system interface in Bengali to ensure accessibility for local farmers posed challenges related to font rendering, layout alignment, and maintaining usability across different screen sizes.
- **Integration Issues:** Combining multiple modules such as crop recommendation, equipment rental, storage booking, and expert consultation into a single platform required careful planning to avoid feature conflicts and maintain system stability.
- **Collecting Free and Reliable APIs:** Finding reliable free APIs for weather and agricultural data was challenging due to limitations in data availability, request quotas, and regional accuracy for Bangladesh.
- **Hardware Limitations:** Limited hardware resources during development and testing affected performance testing and large-scale data processing, requiring optimization of queries and application logic.
- **Scalability Considerations:** Ensuring that the system design remains scalable for future user growth and feature expansion required careful architectural decisions.

Future Enhancements

Although FarmFriend successfully delivers core agricultural support features, several enhancements can be implemented to further improve system functionality, scalability, and real-world impact.

- **Integration of IoT Sensors for Real-Time Field Data:** Future versions of FarmFriend can integrate IoT-based sensors to collect real-time data such as soil moisture, temperature, humidity, and nutrient levels. This data can be directly fed into the system to provide more accurate crop and fertilizer recommendations, enabling precision farming practices.
- **Mobile Application Development:** Developing a dedicated mobile application for Android and iOS platforms would increase accessibility for farmers, especially in rural areas. Offline support and push notifications can further enhance usability in regions with limited internet connectivity.
- **AI-Based Yield Prediction:** Advanced machine learning models can be implemented to predict crop yield based on historical data, weather patterns, soil conditions, and farming practices. This feature would help farmers plan harvests, manage risks, and make informed financial decisions.
- **Multilingual Support:** Adding support for multiple local languages and dialects, including Bengali, would make the platform more accessible to farmers with limited technical literacy. A multilingual interface would significantly improve user adoption across different regions of Bangladesh.
- **Advanced Analytics Dashboard for Farmers:** An analytics dashboard can be developed to visualize farming data such as crop performance, resource usage, rental history, and storage statistics. Graphical insights and reports would enable farmers to track progress, identify trends, and optimize decision-making over time.

Conclusion

FarmFriend successfully demonstrates the application of software engineering principles in solving real-world agricultural problems. By combining Django, React, REST APIs, and automated testing, the system ensures scalability, reliability, and maintainability.

The project highlights the importance of structured development, modular design, and continuous testing. With further enhancements such as IoT integration and AI-based analytics, FarmFriend has strong potential to evolve into a full-scale smart agriculture platform.

About Team ReBuggers

আমাদের টিম

টিম ফার্ম-ফ্রেন্ড

আমরা ৩ জন চেয়েছিলাম একটি প্রোটোটাইপ তৈরি করতে যা কৃষকদের সাহায্য করবে প্রযুক্তির সাথে পরিচিত হতে এবং কৃষিতে উদ্ভাবন যোগ করবে। মাসব্যাপী গবেষণা এবং আপনার সমর্থনের মাধ্যমে, আমরা এই সাফল্য অর্জন করতে সক্ষম হয়েছি।

আমাদের টিমে যোগ দিন



Iftekharul Islam Refat
Full Stack Developer & Automation Enthusiast



S.M. Nabiul Islam
Project Manager & Full Stack Developer



Md. Mezabur Rahman Rasel
Front End Developer & Graphics Designer

Figure 14: Team ReBuggers

আমাদের সম্পর্কে নতুন তথ্যাদি

Champion



May 15, 2025 Nabiul Islam Nabil 10

Champion - Team ReBuggers
won UIU CSE Project Show Spring 2025 for ResQ!

3rd Runner Up



August 20, 2025 Nabiul Islam Nabil 7

3rd Runner Up - Team ReBuggers
at UIU CSE Project Show Summer 2025 for ARAM!

5th Runner Up



December 18, 2023 Nabiul Islam Nabil 3

5th Runner Up - Team ReBuggers
at UIU CSE Project Show Fall 2023 for EasyNeeds



October 11, 2024 Nabiul Islam Nabil 2

Team ReBuggers made a River Analysis Boat
for CSE Project Show Summer 2024

Figure 15: Achievements of Team ReBuggers